PLAN CREATION

From a signed strategy to the printed week the client eats

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Surfaces: Client record · Diet Plans tab · Plan Studio · Preview · Printed PDF · Reference: the Quanutrition / Zoconut tennis plan (26 pages) and the practitioner's plan-creation notes

The strategy says what the body needs. The plan says what the person eats on Monday at 10 a.m. NutriAlign now closes that loop — the same seven-day discipline the practice already prints, with the maths, the guardrails and the sign-off attached.

The week is the product

A nutrition practice does not sell a PDF. It sells a week that a person can follow: a sequence of meals that fits their training, their kitchen and their blood work, printed in a format they already trust. The strategy studio built the clinical decision. Plan Creation builds the week that delivers it, and prints the document the client receives.

The reference for this release is not a wireframe. It is a real 26-page export from Quanutrition's Zoconut account: a tennis player's seven-day plan, meal slot by meal slot, portion by portion, with a recommendations page, 37 numbered general guidelines and a five-paragraph disclaimer. The practitioner's notes alongside it describe the workflow: mandatory plan details first, a strategy dropdown, add-meal and drag-and-drop editing, per-day diet notes, a food picker, a preview, and a PDF download.

This release implements that workflow end to end. Every dish, portion and calorie in the shipped catalogue is parsed from the reference PDF, so the demo plan reproduces the reference numbers exactly (Monday 1,806.6 kcal, Tuesday 3,042.6 kcal). What changes is the machinery underneath: the plan is a structured record with guardrails, not a table typed into a form.

1.1 What ships in v1

- A Diet Plans tab on the client record with plan history, templates, PDFs and revisions.

- A seven-section Plan Studio: Details, Targets, Week plan, Recommendations, Guidelines, Preview, Review & publish.
- A 104-dish food catalogue with per-unit calories, 26 portion units, 10 categories, and manual entries for anything else.
- A seven-day board with meal slots, portions, alternatives ("or"), per-slot meal notes, per-day diet notes, drag-and-drop ordering, copy-day, and a match-target-to-meals action.
- Targets pulled from the published strategy in one click, with an invariant macro split that always totals 100%.
- Diet, allergy and restriction guardrails computed against the catalogue, with an acknowledgement trail.
- A readiness checklist and attestation gate before publishing, plus immutable versions and revisions.
- A server-rendered A4 PDF that mirrors the reference layout: cover, day blocks, recommendations, guidelines, disclaimer.
- Save-as-template so a week built once can be reused on the next client.

1.2 Success measures

Measure	Target	Why it matters
Time to a printable week	< 20 minutes from a published strategy	The Tuesday-afternoon constraint
Reference fidelity	Same portions, calories and section order	Clients compare against the old PDF
Maths integrity	100% of days total 100% macro split	No contradictory numbers in front of a client
Guardrail coverage	Every dish checked against diet, allergies, restrictions	Clinical safety
Sign-off completeness	No plan published without attestation	Defensible record
PDF availability	Inline preview and download, identical bytes	Preview equals what the client gets

What the reference gets right — and what it costs

The Zoconut screens are a competent operating surface. The meal grid is legible, the day tabs carry live calorie and macro totals, the template dropdown respects that a practitioner builds the same shape every week, and the printed plan is calm: an orange band per meal slot, three columns, a note under the dish, and a footer that repeats the practice's contact details on every page.

The cost is hidden in the seams. Names are typed free-hand, so "CUT FRUITS" and "cut fruits" become two foods. Calories are entered per row, so a portion edit silently desynchronises the day total. There is no guardrail against a non-vegetarian dish on a vegetarian client's plan. There is no version trail behind a published PDF. And the macro maths can disagree with itself: a day that says 2.2–2.4 g/kg protein while the split says 10% is a plan a client can catch you on.

2.1 Five failure modes this release removes

Failure mode	What it looked like	What v1 does
Re-typing	Every dish and portion typed by hand, every week	Catalogue picker with 104 dishes, per-unit calories, and manual fallback
Silent desync	Calories edited independently of portions; day totals drift	Portion × per-unit energy is the only source of truth; totals derive
Impossible macros	Protein guardrail changed without the split changing	Editing grams or any percentage rebalances the other two to 100%
Clinical blind spot	Non-vegetarian dish on a vegetarian plan	Diet-tag conflicts, allergy matching, acknowledgement before publish
No trail	A PDF sent and then edited in place	Publish locks the plan, writes a version, revisions fork a new draft

2.2 The design principle

> *The plan is a calculation, not a document. Everything the practitioner sees — the day totals, the target line, the macro bar, the printed calories — is derived from one structure. The PDF is a rendering of that structure, which is why the preview can be the exact file the client receives.*

Who it is for, and the workflow it replaces

The primary user is a clinical nutritionist in a small practice, seeing six to ten clients a day, building most plans from a rotation they already know. Their week is not a blank page; it is a variation on last month's plan, adjusted for training load, travel and the latest bloods.

The secondary user is the client, who only ever sees the PDF and the WhatsApp message that carries it. Everything in the printed document is written for them: the capitalised dish names, the note under each dish, the "or" alternative, the guidelines numbered one to thirty-seven, and the disclaimer.

3.1 Jobs to be done

Job	Current workaround	In the product
Turn a signed strategy into a week	Rebuild the week in Zoconut from a template	New plan → template → targets pulled from the strategy
Adjust for a heavy training day	Edit calories per row and hope totals keep up	Per-day target override or match-target-to-meals
Offer a swap	Type "or" into the dish note	First-class alternatives with their own

		portion and calories
Warn about a restriction	Remember it, or catch it on review	Automatic conflicts with acknowledgement
Send the plan	Export PDF, rename, WhatsApp	Preview, download, published and versioned
Reuse a good week	Ask a colleague for their file	Save as template, available in the New plan menu

End-to-end flow

1. Open the client record → Diet Plans. The history shows the current plan, earlier versions, and two templates.
2. New plan: blank week, the reference-derived sports template, or a template saved from an earlier plan.
3. Details: title, start date, duration, diet type, linked strategy, practice branding. Optional: save this week as a template.
4. Targets: pull from the published strategy, then tune energy, protein grams and the split. Per-day overrides for training days.
5. Week plan: pick a day, add meal slots, add dishes from the catalogue, set quantities and units, add notes, alternatives and day notes. Reorder by drag or with the arrow buttons.
6. Recommendations and Guidelines: select superfoods, add practice notes, choose which of the 37 guidance sections print.
7. Preview: the exact PDF in an iframe. Download produces the same bytes with an attachment disposition.
8. Review & publish: readiness checks, variance and guardrail acknowledgements, attestation, publish. The PDF now carries Plan Approved.
9. Later: open the published plan, download again, or create a revision that forks a new draft without touching the signed version.

4.1 What the practitioner never has to do

- Type a dish name from scratch unless it is genuinely new (manual entry keeps the calories they type).
- Reconcile day totals by hand: calories, protein, carbohydrate and fat are summed from the meals.
- Remember which of the 37 guidelines they include: the selection is stored on the plan and printed in order.
- Rename the PDF or rebuild the cover: client details come from the record, plan details from the plan.

Functional requirements

5.1 Plan shell and history

ID	Requirement	Acceptance
FR-01	Diet Plans tab on the client record	Shows current plan, earlier plans, templates, PDF link per plan
FR-02	Create a plan	Blank week or any template; created as a draft with seven empty days
FR-03	Plan history	Status, version, dishes, kcal/day, target, days built, readiness, updated time
FR-04	Delete a draft	Allowed for drafts only; published plans 409
FR-05	Duplicate	Copies the full week into a new draft
FR-06	Save as template	Stores the week; appears in the New plan menu for any client

5.2 Details

ID	Requirement	Acceptance
FR-07	Title, start date, duration	End date derived from start + duration - 1
FR-08	Diet type	Defaults from the client record; editable
FR-09	Strategy link	Dropdown of the client's strategies; selecting one pulls its targets
FR-10	Branding	Practice name, tagline, email, phone, footer credit used by the PDF
FR-11	Prepared by + internal description	Prepared-by prints; description stays internal

5.3 Targets

ID	Requirement	Acceptance
FR-12	Energy target	Editing calories rescales macro grams from the split
FR-13	Protein target	Editing grams recomputes the protein percentage; carbohydrate absorbs it; fat is preserved
FR-14	Percentage edit	Editing any percentage rebalances the other two so the total is exactly 100
FR-15	Guardrail feedback	Shows g/kg against the strategy range and warns when outside it

FR-16	Per-day override	Any day can carry its own target; reset restores the plan target
FR-17	Match to planned week	Sets the plan target to the average of active days

5.4 Week plan

ID	Requirement	Acceptance
FR-18	Seven-day board	Day tabs carry live calories and a target bar
FR-19	Meal slots	Add from 14 presets or a custom slot; rename, retime, reorder, delete
FR-20	Add dish	Catalogue search and category filter; quantity and unit on the row
FR-21	Manual entry	Name, quantity, unit, calories for that portion; macros stay blank
FR-22	Portions	Quantity × per-unit energy; unit change converts by gram weight
FR-23	Notes	Dish note, recipe, slot meal note, day diet notes
FR-24	Alternatives	Any dish can carry "or" rows with their own portion and calories
FR-25	Drag and drop	Reorder slots and dishes, and move a dish between slots; arrow buttons mirror it
FR-26	Copy day	Copy the selected day to all others, weekdays or weekend
FR-27	Day state	Active / inactive, fast day, clear day

5.5 Recommendations, guidelines, preview, review

ID	Requirement	Acceptance
FR-28	Superfoods	Multi-select with the rationale printed for each
FR-29	Custom notes	Heading + body, printed with the recommendations
FR-30	Printed sections	Measurements guide, disclaimer and the daily target line can be toggled
FR-31	Guidelines	37 sections, individually selectable, printed in order
FR-32	Preview	Server-rendered PDF in an iframe; refresh after edits
FR-33	Download	Same document with attachment disposition

		and a clean filename
FR-34	Readiness	Seven checks with fix links and a score
FR-35	Publish	Attestation required; locks the plan; stamps the PDF; writes a version
FR-36	Revision	Forks a new draft from any published plan; the published PDF is unchanged

The calculation rules

The most expensive defect in the reference workflow is arithmetic that disagrees with itself. Version 1 enforces three rules everywhere — in the strategy modal, in the plan targets, and on the printed page.

6.1 Rule 1 — portions drive calories

A meal stores per-unit energy and macros. Calories are always quantity × per-unit energy. Changing the unit converts the per-unit values by the gram weight of the unit before recalculating, so a dish switched from grams to tablespoons keeps its energy density rather than its number.

6.2 Rule 2 — the split always totals 100

Editing a percentage keeps that point and distributes the remainder across the other two in proportion to their previous values. If both were zero, the remainder is halved. There is no state in which the studio shows a total other than 100.

6.3 Rule 3 — protein grams and protein percentage cannot drift

When protein grams or the g/kg guardrail change, the protein percentage is recomputed from grams and energy, carbohydrate absorbs the difference, and fat is preserved. When energy changes, the protein percentage is recomputed so the gram target stays put. The strategy's live-targets card now reports the average day's calories, which is exactly 4C + 4P + 9F for that split.

Input	Change	Result
Protein target 1.2–1.4 → 2.2–2.4 g/kg at 3,100 kcal	Guardrail	Protein 18% → 38%, carbohydrate 57% → 37%, fat stays 25%
Fat 25% → 35%	Percentage	Protein and carbohydrate shrink proportionally; total stays 100%
Energy 3,000 → 2,600 kcal	Energy	Percentages held, grams rescaled; protein g/kg unchanged
Protein grams 150 → 200	Grams	Protein percentage rises; carbohydrate absorbs; fat preserved

Worked example: 3,100 kcal, protein 3.6–3.8 g/kg at 80 kg → mid 3.7 × 80 = 296 g → 296 × 4 / 3,100 = 38%. Fat stays at 25%, so carbohydrate = 100 − 38 − 25 = 37%. The day now reads 37 / 38 / 25, totals 100%, and the protein achieved line says 294 g at 3.67 g/kg — inside the guardrail.

Guardrails and the publish gate

A plan is a clinical document. The product treats publishing as a signature, not a save. Seven checks decide whether the gate opens, and two of them can be acknowledged rather than fixed when the practitioner has a reason.

Check	Passes when	Blocking
Plan details	Title and start date are set	Yes
Energy and protein targets	Both greater than zero	Yes
Seven-day coverage	Every active day has at least one dish	Yes
Portions	Every meal has a portion string	Yes
Energy within target	Each active day within 20% of target, or variance acknowledged	Yes, acknowledgement removes it
Diet guardrails	No diet, allergy or restriction conflict, or acknowledged	Yes, acknowledgement removes it
Guidance library	At least one superfood or guideline selected	No (warn)

Conflicts are computed from catalogue diet tags — meat, fish, egg, dairy, nuts, gluten — matched against the client's food preference, plus name matching for recorded allergies and restrictions. A vegetarian client loaded with the tennis template produces a conflict list, and the plan cannot publish until the practitioner acknowledges it. The acknowledgement is stored on the plan and re-evaluated on every save.

7.1 What publishing does

- Sets status to published and increments the version.
- Archives any earlier published plan for the client.
- Writes an immutable snapshot to plan_versions with the attestation note.
- Locks the plan: PATCH and DELETE return 409.
- Makes the PDF print Plan Approved with the approved date on the cover.

The printed plan

The PDF is the product the client keeps. It is rendered server-side with ReportLab from the same structure the studio edits, so preview and download are the same bytes.

Pages	Content	Notes
1	Cover: Plan Approved pill, Customer Information (six fields in three columns), Plan Details table, energy and macro strip, seven-day average, allergies and restrictions	Branding from the plan, client details from the record
2–21	One page block per day: orange day title and date, planned/target/difference line, one table per meal slot with Portion and Calories columns, diet notes box	Matches the reference band, row and note treatment
22	Recommendations: superfoods table, standard measurements guide, custom notes	Selection comes from the studio
23–26	General guidelines: 37 numbered sections with paragraphs and bullets	Order is fixed; selection is per plan
27	Disclaimer: five numbered paragraphs	Toggled per plan
Every page	Brand wordmark and tagline, hairline rules, footer with brand, email, phone, credit and page number	All text is selectable

8.1 Typography and colour

Element	Specification
Page	A4 portrait, 34 pt side margins, content width 527 pt
Accent	#F08A4B meal-slot bands; #D9692B section headings; #3F9B57 Plan Approved
Dish row	Helvetica-Bold 8.5 pt in caps, note in 7.5 pt grey, portion and calories 8.3 pt
Guidelines	9.2 pt bold section title, 8.6 pt body with 12.4 pt leading
Footer	7.2 pt, four items distributed across the measure, page number right-aligned above

8.2 Fidelity evidence

The reference plan was parsed from its PDF, not retyped: word coordinates and orange band rectangles reconstruct 7 days, 110 meal slots and 284 dishes. The catalogue's per-unit calories come from those portions,

which is why the seeded plan reproduces Monday at 1,806.6 kcal and Tuesday at 3,042.6 kcal — identical to the source document.

Data model and API

9.1 Tables

Table	Purpose	Key columns
diet_plans	Plan record and current draft/published data	id, client_id, name, status, version, strategy_id, data_json, published_at
plan_versions	Immutable published snapshots	id, plan_id, version, data_json, note, created_at
plan_templates	Saved reusable weeks	id, name, description, days_json, created_at

9.2 Endpoints

Method	Path	Purpose
GET	/api/plan-reference	Guidelines, superfoods, units, presets, branding, templates
GET	/api/plan-catalog	Dishes, categories, units
GET	/api/clients/{cid}/plans	History and templates
POST	/api/clients/{cid}/plans	Create blank or from template
GET	/api/clients/{cid}/plans/{pid}	Plan, client, derived, reference
PATCH	/api/clients/{cid}/plans/{pid}	Update a draft
POST	/api/clients/{cid}/plans/{pid}/publish	Publish with attestation
POST	/api/clients/{cid}/plans/{pid}/revisions	Fork a draft
POST	/api/clients/{cid}/plans/{pid}/duplicate	Copy as a draft
POST	/api/clients/{cid}/plans/{pid}/template	Save the week as a template
DELETE	/api/clients/{cid}/plans/{pid}	Delete a draft
GET	/api/clients/{cid}/plans/{pid}/pdf	Render inline or as a download

9.3 Derived payload

GET returns the plan plus a derived object: per-day calories, protein, carbohydrate, fat, target, difference, percentage and status; week totals; conflicts; warnings; and the readiness score with its checks. The UI never

recomputes anything the server owns — it renders what the server derived, so the studio and the PDF can never disagree.

Verification

This is a built product, not a specification. The evidence below was produced on the same machine that serves the studio.

Layer	Coverage	Result
Backend tests	43 pytest tests: strategy maths and gates, plan catalogue reconciliation, reference calories, target pull, publish gates, lock and revision, vegetarian guardrail, manual meal maths, PDF content, history and duplicate	43 passed
Frontend	TypeScript project build and Vite production build	Clean
End-to-end	Playwright: every client tab renders with a filled record; plan history; seven-day board; food picker search and add; manual entry; macro rebalancing; preview PDF fetch; publish checklist	35 of 35 checks
Preview fidelity	PDF fetched from the endpoint and verified as %PDF with 27 pages	Pass
Accessibility	axe-core WCAG 2.1 AA on plan history, week board, targets, review, food picker, mobile	0 violations
PDF text	Cover, day block, diet notes, superfoods, guidelines, disclaimer are real text	Pass

10.1 The test client

A fully-filled record drives the verification: Jaideep Singh, 19, male, 80.1 kg, 180.3 cm, A+, non-vegetarian with a lactose intolerance, tennis player training twice a day, a published strategy at 3,100 kcal and 1.6–1.8 g/kg, and a published 180-day plan built from the reference template with a lighter Monday target. The script is scripts/seed_test_client.py; every tab of the record is populated.

Release notes, limits and what comes next

11.1 Deliberate limits

- The catalogue ships 104 dishes. Anything else is a manual entry with its own calories; macros stay blank rather than guessed.
- Macro grams are estimates from a composition table, scaled so the energy reconciles with the calorie column. The studio says so.
- Templates snapshot the week, not the target maths; targets are re-pulled from the strategy on the new plan.
- A multi-week program reuses the same seven-day rotation; duration only drives the printed period.
- Branding is per plan, not per practice; a settings surface is the natural next step.

11.2 What comes next

Candidate	Value	Note
Practice settings	Set branding once for every PDF	Small, high polish
Grocery list	Aggregate the week into a shopping list	Derives from the same structure
Client sharing	Signed link or WhatsApp send	Needs an auth story
Plan comparison	Strategy v1 vs v2 side by side	Version data already exists
Photo library	Dish images in the client PDF	Catalogue has the hooks

11.3 Definition of done for this release

- A practitioner can go from a published strategy to a printed, approved PDF without touching a spreadsheet.
- Every number on screen and on paper agrees with every other number.
- A vegetarian client cannot be sent a chicken curry by accident.
- The preview is the file, and the file is versioned.
- The client receives a document that looks like the one the practice already sends — and is measurably more correct.

Plan Creation is the moment NutriAlign stops being a record system with a calculator and becomes the surface a practice actually works in. The strategy is the judgement. The plan is the delivery. The PDF is the promise.